Raj Kolala

San Jose, CA | 925-984-8865 | rajkolala10@gmail.com | linkedin.com/in/rajkolala | github.com/RajKolala | Portfolio

Education

San Jose State University **May 2027**

Bachelor of Engineering, Computer Networking System Management, Minor in Business Administration *San Jose, CA*

Relevant Coursework: Network Administration, Python, Analog and Digital Circuits, IoT, Calculus, Linear Algebra, Public Speaking, Business Statistics, Machine Learning Technology and Applications, Network Security & Prevention Management

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, Bash, PowerShell · **Backend:** Node.js, Express, FastAPI, Flask, REST APIs, WebSocket, PostgreSQL, MongoDB, Firebase · **Frontend:** React, React Native, Expo · **AI/ML:** LLMs, RAG, LangChain, embeddings, vector search, Pandas, scikit-learn · **Cloud & DevOps:** AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Linux · **Networking & Security:** TCP/IP, DNS, DHCP, VLAN, VPN, SSH, Wireshark, ISO 27001

Experience

Jabil **August 2026 – Present**

AI & Machine Learning Intern *San Jose, CA*

- Designing **AI copilots for printed circuit board assembly (PCBA)** that ingest a customer's **ODB++ design package** and automatically generate manufacturing outputs including solder stencil files, component sourcing lists, design reviews, and process flow assessments.
- Building copilot solutions on **LLMs and RAG using LangChain**, grounding models in **IPC standards, component libraries, and past designs** to produce outputs that are accurate, explainable, and manufacturable.
- Constructing the **retrieval knowledge base and data layer** by parsing and normalizing **ODB++ files with embeddings and vector search**, transforming raw design packages into structured inputs the models can reason over.

Elide **May 2025 – Aug 2025**

Network Systems & Infrastructure Intern *San Jose, CA*

- Deployed a **brand new office network** for a 20+ person startup from the ground up, handling end-to-end **physical cabling, switch and router configuration**, and **wireless access point setup** across the entire facility.
- Designed and implemented **VLAN segmentation** to separate executive and leadership traffic from general staff, enforcing **role-based network access control** across the organization.
- Configured a **guest and event network** with **temporary access control** to support visiting engineers and company-hosted events, isolating external traffic from internal systems.

Cisco **July 2023 – Nov 2023**

Network Engineering Consultant *Santa Clara, CA*

- Collaborated with **150+ IT professionals and educators** to deliver workshops on **data literacy, cybersecurity, and generative AI integration**, configuring practical solutions for real-world business use cases.
- Designed and facilitated breakout sessions on **AI-driven productivity tools** and **cybersecurity principles**, achieving a **92% satisfaction rate** across **20+ organizations** while supporting consulting sessions diagnosing **data anomalies and security vulnerabilities** across distributed enterprise infrastructures.

Projects

Clarity – See Yourself | *React Native, Expo, WebSocket, Node.js, EAS Build* **Mar 2026 – Present**

- Architected a **real-time peer-to-peer video mirroring system** for iOS using **WebSocket protocol** over a **Node.js relay server**, enabling sub-second latency camera feed synchronization between two devices via a **6-character room code session handshake**.
- Developed and shipped a **full-stack iOS application** to the App Store integrating a **client-server WebSocket architecture**, live **camera stream processing** with 14 real-time editing parameters, and local on-device storage with **zero data transmission to external servers**. Companion website live at clarityseeyourself.net.

Cardiovascular Disease Classification & Prediction | *Python, Pandas, scikit-learn* **Mar 2026 – Apr 2026**

- Built a **binary ML classifier** to predict cardiovascular disease across **70,000 patient records**; performed end-to-end data preparation including **outlier removal**, **age conversion from days to years**, **StandardScaler normalization**, and correlation analysis using Pandas.
- Trained and compared **5 classification models**; applied **RandomizedSearchCV with cross-validation**, achieving **73.7% accuracy (Random Forest)**; evaluated on precision, recall, and F1 with zero overfitting confirmed via train/test gap analysis.

End-to-End Network Prototype | *Cisco Packet Tracer, TCP/IP, Routing, DNS* **Oct 2025 – Dec 2025**

- Architected and simulated an internet-scale, **multi-tier ISP topology** connecting **home, SMB, and cellular access networks** to a centralized data center through hierarchical upstream transit.
- Engineered **Tier-3, Tier-2, and Tier-1 ISP interconnections**, implementing end-to-end IP addressing, routing, and DNS to validate packet traversal and scalability across **15+ routed networks**.

Certifications

Google Cloud Data Analytics Certificate (Sep 2026) · Cisco: CyberOps Associate (May 2025) · Google: Cybersecurity Professional Certificate (Nov 2025) · AWS Academy: Cloud Architecting (Nov 2025) · CCNA: Enterprise Networking, Security, and Automation (Mar 2025) · CCNA: Switching, Routing, and Wireless Essentials (Jun 2024)